

Stage 1: Egocentric Video Collection

Ego-Centric Video Source

Public Video Data



Dataset: Ego4D

Self-Collected Video Data



Capture Device: Smart Glasses, Smart Phones

Video Collection Standards

- Distinct Spatial Semantics
- Clear Temporal Action Evolution Logic
- Optimal Contextual Duration for Interaction

Covered Scenarios



Kitchen



Restaurant



Retail



Order

Stage 2: Tool Library & Database Construction

Tools Design



Scenario Specificity



Comprehensive Functionality



Parameters Fuzzy Matching

Database Construction



1. Egocentric Video Collection

MENU		
STEAK ... \$25		
PASTA ... \$18		
DISH NAME	PRICE	INGREDIENTS
[STEAK]	[25]	[BEEF, HERBS]
[PASTA]	[18]	[FLOUR, SAUCE]
...	...	...

2. Visual Information Extraction

DISH NAME	INGREDIENTS	DISH NAME	NUTRITION INFO
[STEAK]	[BEEF, HERBS]	[STEAK]	[650 kcal, 45g Pro]
[PASTA]	[FLOUR, SAUCE]	[PASTA]	[400 kcal, 15g Pro]
...	...	...	...

3. External Data Enrichment

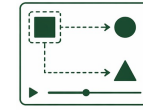
DISH_ID	DISH NAME	INGREDIENTS LIST	NUTRITION DETAILS
...	...	...	...

4. Final Database Construction

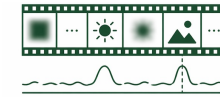
Iterative Co-design

Stage 3: Task Design & Ground Truth Annotation

Multimodal Perception



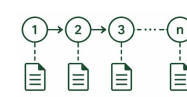
Spatial / Temporal Referential Dependency



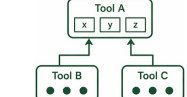
Multimodal Information Sparsity & Dynamics

Task Design Complexity

Logic Reasoning

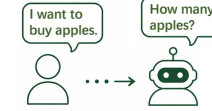


Multi-Hop Logical Dependency



Inter-Tool Parameter Dependency

Interaction



Intent Incompleteness & Active Elicitation



Information Redundancy & Key Extraction

GT Annotation

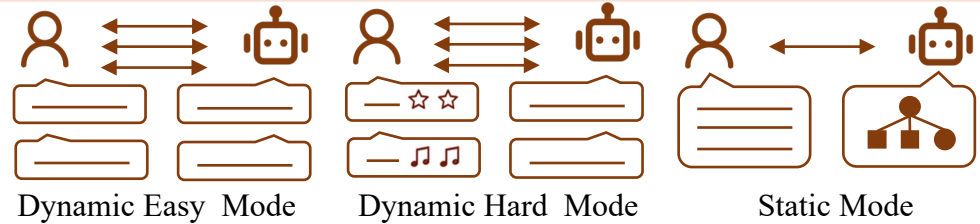
- Multimodal Reference
- Key Tool-call Trajectory



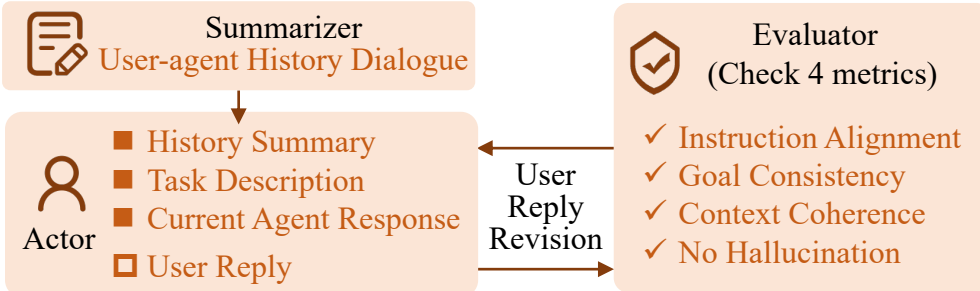
100% Manual Annotation with Quality Review

Multi-agent Simulated User

Three User-Agent Interaction Modes



Simulated User (Multi-agent System)



Joint Validation Framework

Process-based Validation

τ^* (GT): $T_1^* \rightarrow T_2^* \rightarrow \dots \rightarrow T_n^*$

τ (Agent): $T_1 \rightarrow T_2 \rightarrow \dots \rightarrow T_m$

Success if $\tau^* \subseteq \tau$
(covers all necessary tool calls)

Compare Tool-call Sequence τ With Ground Truth τ^*

AND

Result-based Validation

Execute τ^*

Database State S^*

Hash(S^*)

Execute τ

Database State S

Hash(S)

Success if
Hash(S^*) = Hash(S)

Compare Final Database States via Hashing